Acoustic and Perceptual Studies on the Tones of the *Yancheng* Dialect

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Abstract

The *Yancheng* dialect is one of the dialects in the Hongchao part of Jianghuai Mandarin. In this paper, acoustic and perceptual studies of the tones were conducted to investigate tone values, contours, range and class of the *Yancheng* dialect. Results of the acoustic study show that, tone values of the five tones are: T1 (Yinping)[431], T2 (Yangping)[314], T3 (Shangsheng)[443], T4 (Qusheng)[235], and T5 (Rusheng)[4]; in addition, tone contours, which are difficult to reveal in conventional dialectology, are described and explained. Results of the perceptual study on show that T3 of the *Yancheng* dialect exists a categorical perception, in which convex, rising and concave contour tones are included.

1. Introduction

Yancheng is the largest city in Jiangsu Province with a long coastline, and the *Yancheng* dialect belongs to the Hongchao part of Jianghuai Mandarin. Due to its special location as it is far from the center of Jianghuai Mandarin area, the *Yancheng* dialect possesses certain particular characteristics. Generally, the *Yancheng* dialect refers to the dialect spoken by the *Yancheng* urban district and its surrounding area[1].

Liu (2007) maintains that the Hongchao part of Jianghuai Mandarin has five tones, namely T1 (Yinping), T2 (Yangping), T3 (Shangsheng), T4 (Qusheng), and T5 (Rusheng). There are several former studies leaving valuable descriptions about the five tones of the *Yancheng* dialect.

The first record of the tones of the *Yancheng* dialect derives from *An Overview on the Jiangsu and Shanghai Dialects* (1960), in which the tone values are recorded as T1(Yinping) [31], T2(Yangping) [213], T3(Shangsheng) [53], T4(Qusheng) [35], and T5(Rusheng) [5].

Su (1992) reports the tone values of the tones of the Yancheng dialect when comparing it with the *Beijing* dialect. His results show that in the *Yancheng* dialect, five tones are T1(Yinping) [31], T2(Yangping) [213], T3(Shangsheng) [54], T4(Qusheng) [35], and T5(Rusheng) [5]. However, Su (1993) himself changed the value of Shangsheng into [53] along a year later, with an explanation that T3(Shangsheng) begins as high and level, but ends with a little falling.

[1] Jianyan section, which is mentioned in *Jiangsu Chorography*, is consistent with the *Yancheng* dialect.

Gu (1993) reports his investigation in the tone are: T1(Yinping) [31], T2(Yangping) [214], T3(Shangsheng) [53], T4(Qusheng) [35], and T5(Rusheng) [5]. Jiangsu Chorography (1998) also reports a tones inventory of the *Yancheng* dialect, shown below in Table 1.

Cao (1998) first employs experimental study to investigate the tones of the *Yancheng* dialect according to T-value method[2] as shown in Table 2. Liu (2007) also conducts experiments and adopts z-score method[3] to investigate them and gets her results as shown in Table 3.

Although former studies have done certain explorations and investigations in the tones of the *Yancheng* dialect, there are still two points worth noting. First, most studies do not breakthrough limitations of conventional dialectology, depending on former listening and transcriptions. Second, former studies lack certain descriptions and discussions on tone contours, and some even only include experimental records.

Table 1 The tone values of the *Yancheng* dialect in *Jiangsu Chorography* (1998).

Class	T1	T2	T3	T4	T5
Tone Value	31	213	55	35	5

Table 2 Experimental results of the tone values of the *Yancheng* dialect in Cao (1998).

Class	T1	T2	T3	T4	T5
Tone Value	31	213	55	35	5

Table 3 Experimental results of the tone values of the *Yancheng* dialect in Liu (2007).

Class	T1	T2	T3	T4	T5
Tone Value (Male)	41	113	443	24	5
Tone Value (Female)	51	114	554	24	5

2. Experiment 1: Acoustic study

It is generally believed that the maximum number of tone levels any language can distinguish is five (Wang 1967; Maddieson 1978). Chao (1930) first categorizes pitches of tones into five points, and invents a clear and concise way of writing syllabic pitch shapes, the tone letters with numerical values on five-point scale. Such a method is used by conventional Chinese dialectology until now.

Liu (1924: 19) claims that if we consider pitch and length at the same time and use these two parameters to draw a curve to show their functional relation, and we could call this curve as a standard representative of the tone.

Zhou (1993) claims a new approach to investigate phonetics, which is a combination of acoustic and perceptual studies. It employs acoustic statistics to make up for the deficiency of listening, and perceptual results to break instruments' limitations. Combining valid acoustic statistics and tradition tones' descriptions, an

[2] T-value method is proposed by Shi (1990).
[3] Z-score method is proposed by Zhu (2004).

acoustic study is conducted to investigate the tones of the *Yancheng* dialect.

2.1. Method

2.1.1. Stimulus materials

Table 4 above is the stimulus words used in the acoustic study. Wu and Lin (1989: 185) claims that, properties of voiced or voiceless of plosives before vowels along with vowels' tone quality would bring influences on fundamental frequencies (F0) of tones. Therefore, according to the collocation of consonants, vowels and tones (Su 1993), we design three sets of common Chinese characters, namely zero initials with high vowel finals (衣 set), voiceless initials with high vowel finals (潽 set), and voiced initials with low vowel finals (妈 set)[4].

Table 4 Stimulus words used in the acoustic study.

Long tones				Short tone
Yinping	Yangping	Shangsheng	Qusheng	Rusheng
衣 (clothes)	移 (to move)	椅 (chair)	意 (meaning)	一 (one)
潽 (boiling)	蒲 (calamus)	普 (common)	铺 (bed)	瀑 (fall)
妈 (mother)	麻 (numb)	马 (horse)	骂 (scold)	抹 (smear)

2.1.2. Participants

Four speakers (2 females and 2 males) were recorded. All were local speakers who have no known history of immigration and speech impairment.

Table 5 Participants' information.

Speakers	Gender	Age	History of living outside
F1	Female	20	No
F2	Female	71	No
M1	Male	47	No
M2	Male	22	Yes

2.1.3. Recordings

We used Praat[5] to record mono sounds with sampling frequency of 44100Hz and quantification precision of 16bt. Recording were conducted in silent rooms, and speakers were required to read characters in Table 5 in front of the microphone. Each

[4] Other three sets (zero initials with low vowel finals, voiceless initials with low vowel finals, and voiced initials with high vowel finals) do not have corresponding characters in the *Yancheng* dialect.

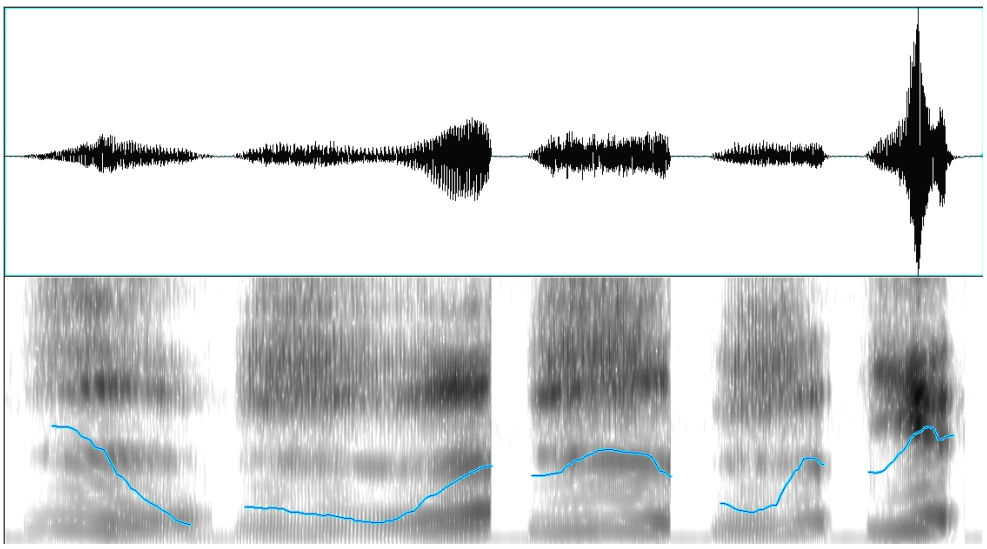
[5] Praat, http://www.fon.hum.uva.nl/praat

character was read four times separately, and we chose the best record. After recording, we saved each file as a format of .wav, and categorized each speaker's records in different folders.

2.2. Acoustic data processing

The first step of acoustic data processing is to extract F0 from sound files. As shown in Figure 1, sounds of 衣 set from M2 are manually cut and put together. The upper half shows the waveforms, while the lower half shows the spectrograms, in which the blue lines are pitch lines of F0.

Figure 1 The sounds from M2, 衣、移、椅、意、一, in the *Yancheng* dialect.



It is believed that, the starting point of a tone begins from that of a vowel, which is from the second impulse of a vowel in the waveforms (Lisker & Abramson 1963: 416; Baken 1987: 376). And according to the clearness of F2, we decide the final point. Based on the rules set above, we cut each tone segment. After measuring the durations, we begin to process the segments. Long tones are segmented by 11 equal diversion points as sample points, while the short by 6, to extract F0 from each point. Then, we figure out the mean values of F0 on each sample point of each tone[6].

We employ z-score approach[7] to normalize tones, using Excel to make four z-score figures of tones of each speaker[8]. Meanwhile, according to the standard

[6] See full details in Appendix Table 1~4.

[7] Z-score method is proposed by Zhu (2004), which deals with random differences in tones. It employs overall measurements to decide the frequency domain, and overcomes the deficiency of distorting tones' nature of T value method. (Shi 1990). The formula of z-score method is $z_i = \frac{x_i - m}{s}$, in which x_i is an observed value, m is a mean value of a given sample, and s is a standard deviation.

[8] See full details in Appendix Figure 1~4.

deviations of four speakers, we figures out their deviation means, and a z-score figure of tones of the *Yancheng* dialect is made, as shown in Figure 2 below.

In order to compare this study results with former studies, we transform the z-score values to numerical values on five-point scale. According to the distribution of z-score values, we stipulate the range as [-2, 2][9]. For long tones, we set three points, 0%, 50%, and 100%, as sample points; for short tones, we set two points, 0% and 100%, as sample points. Then we transform z-score value to numerical value on five-point scale as shown in Table 6 below.

Figure 2 The average z-score value diagram of the tones of the *Yancheng* dialect.

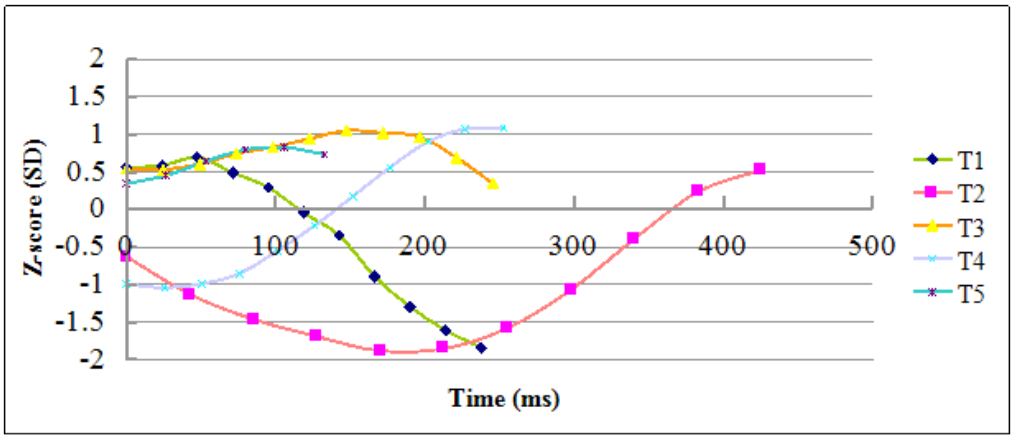


Table 6 The transformed results in the tone letter form of the tones of the *Yancheng* dialect.

Class	T1	T2	T3	T4	T5
Tone Value	431	314	443	235	4

2.3. Results

We compare the results with records and descriptions from Su (1993), as shown in Table 7 below.

[9] Zhu (2004) describes a transform approach for frequency domains called POR. A mathematical relationship, $g=\frac{x-(m-cs)}{(m+cs)-(m-cs)}=\frac{x-m+cs}{2cs}=\frac{1}{2c}\cdot\frac{x-m}{s}+\frac{1}{2}=\frac{1}{2c}z+\frac{1}{2}$, exists between POR and z-score. In addition, following the rules of POR, the upper limit value of the frequency domain is 1, while the lower of that is 0. If the value lower than 0, we set the value of g as 1, and if the value higher than 1, we set the value of g as 5. Therefore, we can get simultaneous equations from limitations of the rules and that mathematical relationship, and figure out that $-c\le g\le c$, which means, the upper limit value is c while the lower limit value is -c.

Table 7 A comparison between the results and the records from Su (1993) of the tones of the *Yancheng* dialect.

Class	T1	T2	T3	T4	T5
Experiment 1	431	314	443	235	4
Su (1993)	31	213	53	35	55

We can discover from Figure 2 in the above part, in the *Yancheng* dialect: T1 (Yinping) is a high falling contour; T2 (Yangping) is a concave contour; T3 (Shangsheng) is a convex contour; T4 (Qusheng) is a rising contour; T5 (Rusheng) is a high short contour.

T1 (Yinping): The tone value [431] in our experiment is 1 point higher than [31] in Su (1993). According to Figure 2, we can find from the beginning to the point of about 50 ms, there shows a slight rising contour, and then it falls to the end. On the whole, the basic tone contour is comparatively consistent with Su's record, but a difference from pitch span exists.

T2 (Yangping): The tone value [314] in our experiment is different from [213] in Su (1993). Through observations on each speaker's tone value, we find that, the basic tone contour keeps comparatively similar, though differences exist in tone values. We think, as Yangping is the only concave contour in the tone pattern of the *Yancheng* dialect, those differences could be indistinctive.

T4 (Qusheng) and T5 (Rusheng): The tone value of Qusheng [235] in our experiment is 1 point lower than [35] in Su (1993), but the contour seems similar as a rising one, which is consistent with former studies. The tone value of Rusheng [4], compared with records of Su (1993) and other studies, is 1 point lower, and meanwhile the tone contour exist a rising tendency.

T3 (Shangsheng): The tone value is [443], which is the same as male tone values record in Liu (2007). However, there are great differences between ours and others'. It's discovered in our experiment that, the tone value is 1 point lower than conventional studies. Meanwhile, at a perspective of tone contour, T3 of the *Yancheng* dialect is neither a level pitch nor a fall one. Instead, it is a convex contour. Yip (1980) reports her tone system, in which two convex-contour tones are included in twelve tones, described as [+upper, LHL] and [-upper, LHL]. In her analysis in Fuzhou dialect, the tone 242 is represented as [+upper, LHL] (Yip 1980: 341), and she allows sequences of three tone features[10]. Wang (1967) claims that tones have seven phonological feature, namely [± contour], [± high], [± central], [± mid], [± rising], [± falling], and [± convex]. Based on our experimental data, we consider, T3 of the *Yancheng* dialect is also a convex-contour tone as mentioned in the tone systems of Yip (1980) and Wang (1967).

2.4. Discussions

As we look at z-score value diagrams of each speaker, we find a noticeable

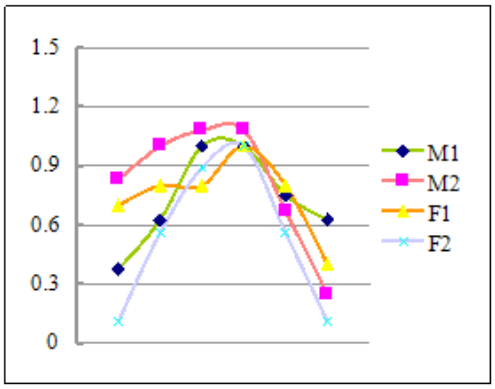
[10] Bao, Z. (1990) *On the Nature of the Tone*. Doctoral dissertation, MIT, Cambridge, MA.

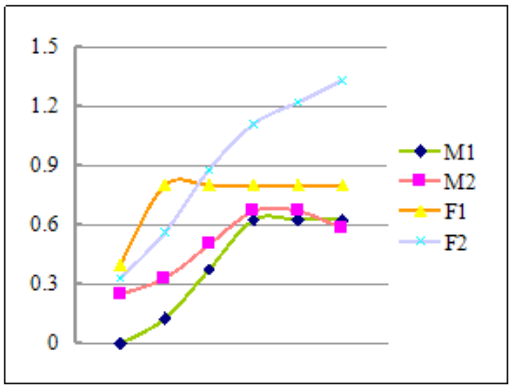
phenomenon in our experiment. That is, F2's tone values and contours are quite different from those of others, though they keep a systematic consistence. Speaker F2 is a 71-year-old local woman, and the age gap between her and others is obvious. We consider her pronunciation should be a representative of the old *Yancheng* dialect.

We compare and analyze the most obvious differences in T3 and T5 between her and others, as shown in Figure 3 and Figure 4 below. Through comparison via the figures, we find that: first, though F2's F0 is not the highest, her degree of convexity is obviously higher than other three; second, though Rusheng sounds quite short, F2's pitch increase value is quite higher than other three shown as a steep slope.

Therefore, we surmise a possible evolution of tone contours of T3 and T5 of the *Yancheng* dialect, which is a tendency of decrease in degree of convexity or rising. However, due to limitations of speakers and data, we now could not go further and draw such a conclusion, but it is still a good open question.

Figure 3[11] Curves of Shangsheng contours from each speaker (Left).
Figure 4 Curves of Rusheng contours from each speaker (Right).





3. Experiment 2: Perceptual study on T3 (Shangsheng)

In the acoustic study, we find that, T3 of the *Yancheng* dialect is not consistent with some former studies, and shows clear phonological features of [+convex] (Wang 1967). We suppose, in the *Yancheng* dialect, distinctiveness of the degree of convexity of T3 exists in perception. Theoretically, for *Yancheng* people, when the convex contour rises or falls to a certain limit, it would be no longer perceived as T3. In order to confirm this supposition, we design a perceptual study.

3.1. Methods

The perceptual study employs categorical perception (Liberman et al., 1957) to investigate T3 of the *Yancheng* dialect.

[11] The y-axis represents z-score value, and its unit is standard deviation, and the x-axis employs comparative duration. The same as right.

3.1.1. Stimuli

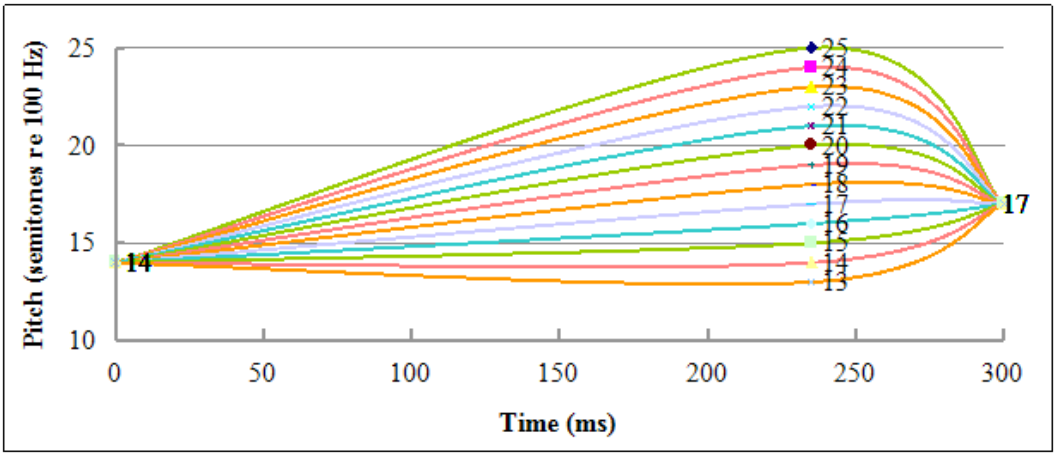
As continuous acoustic changes evoke categorical perceptual effects for tones, we select "椅 (chair)" sound from F2 in the acoustic study as the raw sound material, and fix the length and the starting and final points. The pitch of the starting point is 15 semitones[12], while that of the final is 17 st. The total length is 300ms. We extract the maximum pitch in the raw sound material, which is 21 st, as an inflection point, and the point is situated in 235 ms.

Morest (1965) reports that, pitch class is a tonal quality which recurs among tones which share the relationship of an octave. Wright et al. (2000) points out, while boundaries of musical pitch categories vary among human cultures, the recognition of octave relationships is a natural characteristic of the mammalian auditory system. Therefore, we consider common people's perceptual tonal range should be in an octave, which is 12 st. According to the raw sound material, we synthesize thirteen different pitch stimuli via PSOLA in Praat, and each of them has different contours as shown in Table 8 and Figure 5 below. Each stimulus has a distance of 1 st at the inflection point. The highest F0 is 25 st while the lowest is 13 st, and as it turns out, the difference is an octave.

Table 8 A scale of synthesized stimuli of T3 of the *Yancheng* dialect.

Stimulus	1	2	3	4	5	6	7	8	9	10	11	12	13
Pitch at the inflection point	13	14	15	16	17	18	19	20	21	22	23	24	25

Figure 5 Acoustic continua of synthesized stimuli of T3 of the *Yancheng* dialect.



3.1.2. Participants

Ten local people (6 females and 4 males) participated in the perception experiment. All were born and lived in *Yancheng*. None reported any hearing impairment.

[12] Semitones re 100Hz, and we get round numbers. We use "st" for short below.

3.1.3. Procedure

Thirteen synthesized stimuli were played via Praat. Participants were tested individually in a quiet room, listening to a random sequence of stimuli.

In the identification test, we employ Forced-choice Paradigm method to require participants judge whether a certain stimulus they listen is "椅 (chair)". There is a 500 ms mute interval between each stimulus. The whole test is repeated for six times.

In the discrimination test, we employ Same-Different method to require participants to discriminate whether two adjacent stimuli played in a pair are the same. The short mute interval between them is 300 ms, and a long mute interval in 500 ms exists between each pair. The whole test is repeated for eight times.

3.2. Results

Figure 6 Results of the identification test of T3 of the *Yancheng* dialect.

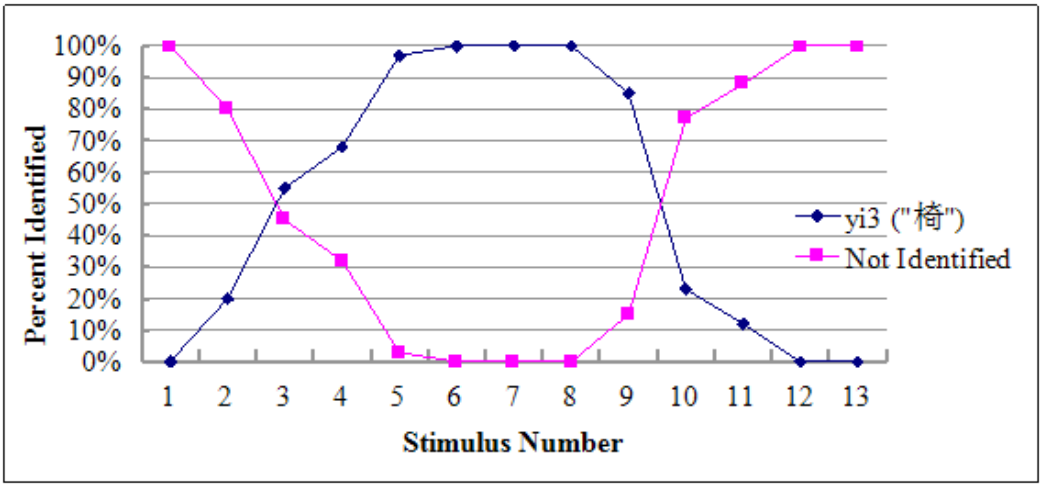


Figure 6 above is *Yancheng* people's perceptual effects in the identification test on T3. We can find from the curve in Figure 6 that, *Yancheng* people begin to perceive T3 match with "椅 (chair)" from Stimulus 3, and end perceiving it to Stimulus 9. However, Stimulus 3 and 4 should not be included as identified stimuli, since they have no significantly statistical difference according to the test of significance[13]. Therefore, we consider stimuli identified are Stimulus 5, 6, 7, 8, and 9. Particularly, for Stimulus 5 to 8, perception of identifying them reveal a categorical consistency[14].

Figure 7 below is *Yancheng* people's perceptual effects in the discrimination test on T3. Suppose the standard of effective discrimination is over 50% discriminated, then we can find from the curve in Figure 7 that, *Yancheng* people can discriminate Stimulus 3 and 4 with a percent discriminated over 60%. That is a feature of

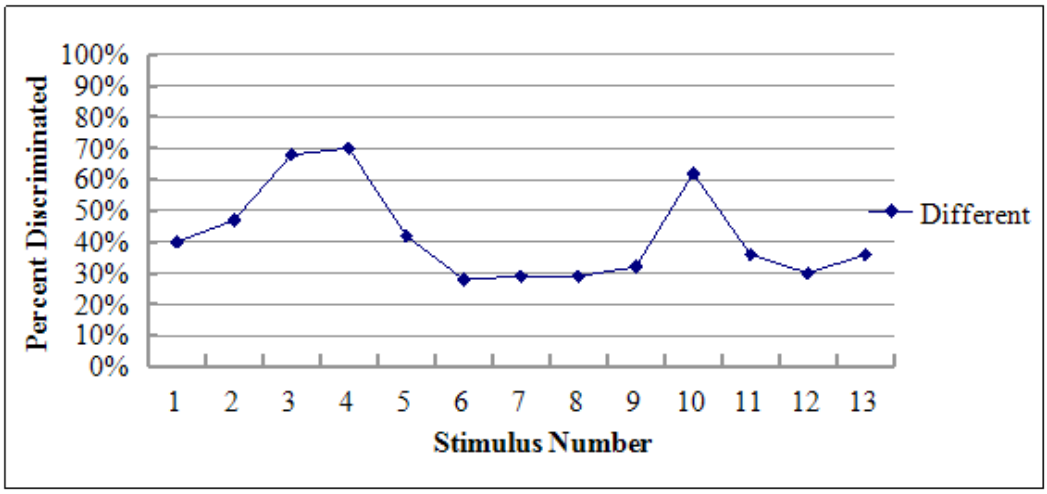
[13] According to the test of significance, Stimulus 3: $\chi^2_{15.st}(1) = 0.6000, \quad P = 0.439$; Stimulus 4: $\chi^2_{16.st}(1) = 7.7760, \quad P = 0.005$.

[14] Percent identified of Stimulus 5 is 97%, which is over 95%. In this case, we consider it as consistent with Stimulus 6, 7, and 8.

continuous perception at the bound of T3. Meanwhile, at Stimulus 10, we find *Yancheng* people also show a strong sense of perception there with a 62% percent discriminated.

In sum, the categorical perception of T3 for *Yancheng* people is between Stimulus 5 and 9 (17 st ~ 21 st), which has a tonal range of 4 st.

Figure 7 Results of the discrimination test of T3 of the *Yancheng* dialect.



3.3. Discussions

It is noticeable that, though we prove T3 a convex contour in the acoustic study, we also find Stimulus 10, 11, 12, and 13 are not perceived as T3, especially as Stimulus 12 and 13 with a 0% percent identified. It seems that no all convex contours would be perceived as T3. Thus, we consider convex contours and Shangsheng ones are not exactly corresponding to each other. There is a upper limit of T3 of the *Yancheng* dialect, which excludes some unnatural and highly convex contour tones.

Furthermore, another noticeable phenomenon is that, the perceptual category of T3 includes convex tones as Stimulus 5 and 9. Meanwhile, as we see Stimulus 4 as a rising tone, this suggests that there is a lower limit of T3 of the *Yancheng* dialect as well, which is very close to a rising tone without convexities.

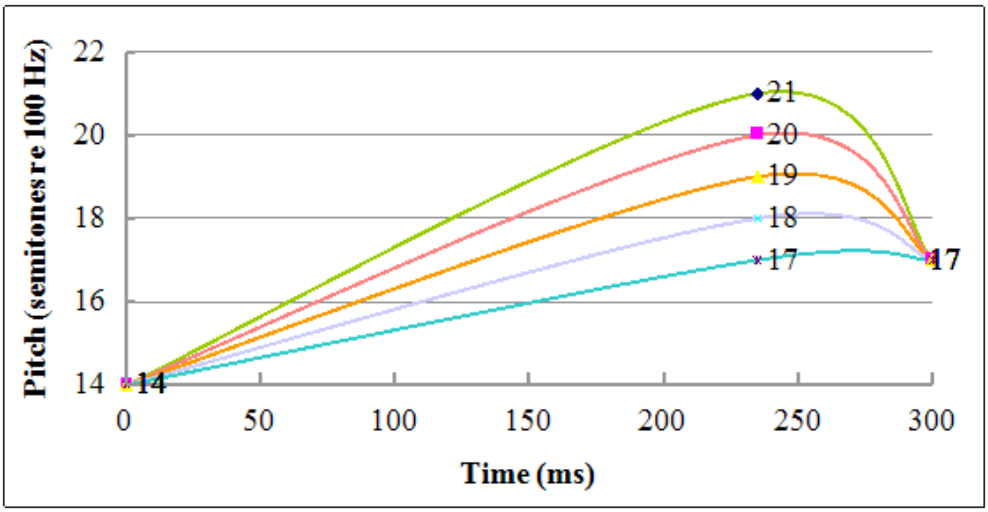
4. Conclusion

Table 9 The tone inventory of the *Yancheng* dialect.

	Yinping	Yangping	Shangsheng	Qusheng	Rusheng
	431	314	443	235	4
high	+	+	+	+	+
rising	−	+	+	+	−
convex	−	−	+	−	−
concave	−	+	−	−	−
short	−	−	−	−	+

We employ acoustic and perceptual studies to conduct a comprehensive research in the tone values, contours, range and class of the *Yancheng* dialect. The acoustic study shows the tone system of the *Yancheng* dialect. We use Chao's (1930) numerical values on five-point scale and Wang's (1967) phonological features of tones, also plus distinctive features [± concave] and [± short] added by ourselves, to show the tone inventory of the *Yancheng* dialect in Table 9 above. The perceptual study shows *Yancheng* people's perceptual category of T3, which is a crescent area with a 4 st tonal range, as shown in Figure 8.

Figure 8 The perceptual category of T3(Shangsheng) of the *Yancheng* dialect.



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Appendix:

Appendix Table 1 M1's F0 values of the tones of the *Yancheng* dialect (Hz).

Hz	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Yinping	184	184	189	184	174	159	148	137	128	120	115
Yangping	134	129	123	117	113	115	125	142	162	180	178
Shangsheng	180	175	175	180	186	194	197	195	190	187	181
Qusheng	126	126	130	136	148	160	172	189	202	206	208
Rusheng	163		166		173		183		184		183

Appendix Table 2 M2's F0 values of the tones of the *Yancheng* dialect (Hz).

Hz	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Yinping	240	241	244	232	217	195	173	154	137	125	118
Yangping	133	127	117	111	112	112	117	128	151	171	207
Shangsheng	227	228	232	238	241	245	245	243	238	221	197
Qusheng	135	133	134	138	145	156	176	195	213	223	215
Rusheng	193		198		210		220		220		214

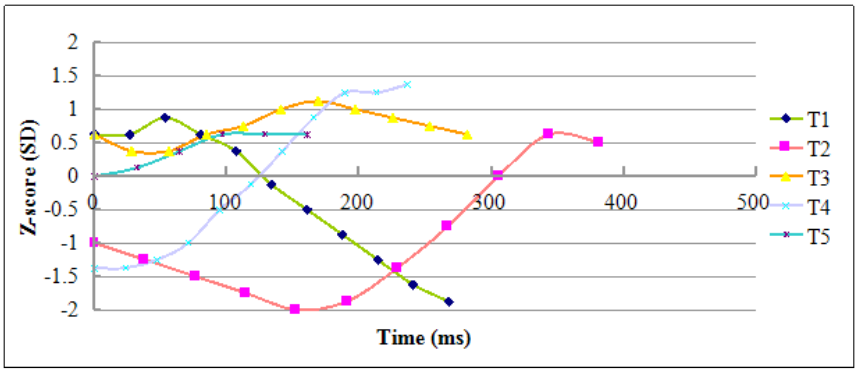
Appendix Table 3 F1's F0 values of the tones of the *Yancheng* dialect (Hz).

Hz	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Yinping	232	238	242	230	221	213	204	193	188	183	178
Yangping	224	210	194	185	176	178	180	188	200	220	229
Shangsheng	240	244	245	246	246	248	249	250	249	244	233
Qusheng	213	212	212	215	218	227	235	245	253	261	261
Rusheng	248		248		245		244		243		237

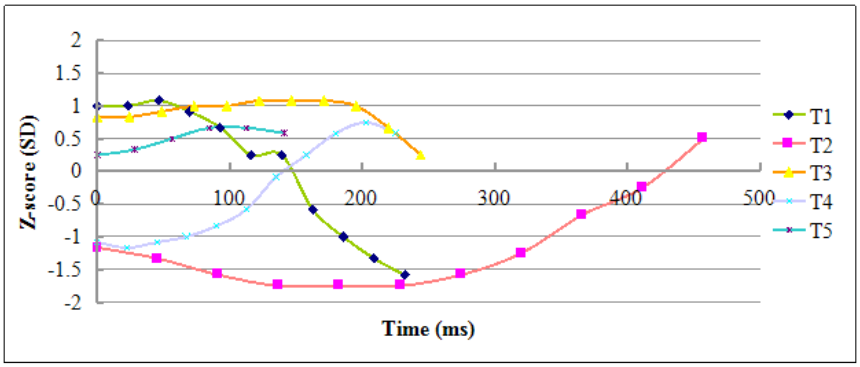
Appendix Table 4 F2's F0 values of the tones of the *Yancheng* dialect (Hz).

Hz	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Yinping	250	243	249	252	258	247	225	198	180	170	162
Yangping	222	184	173	168	165	165	172	210	245	280	292
Shangsheng	245	246	256	269	282	290	292	297	296	272	246
Qusheng	192	187	187	196	209	226	242	259	279	291	298
Rusheng	256		270		289		300		308		316

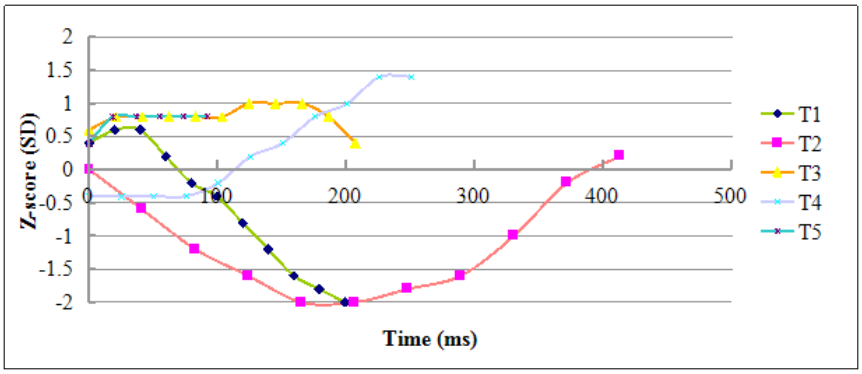
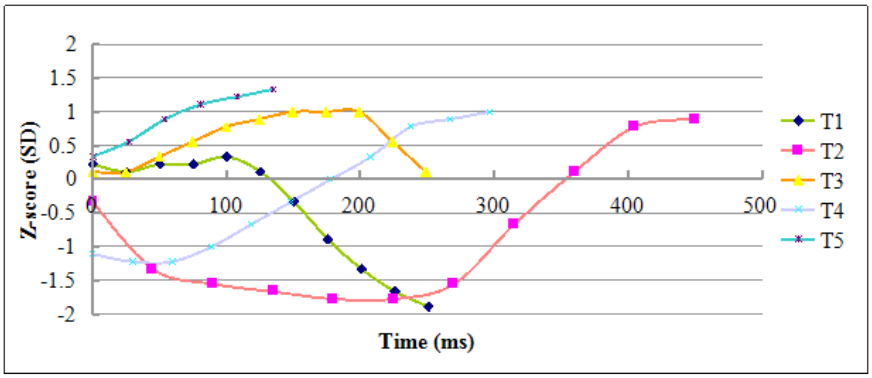
Appendix Figure 1 M1's z-score values diagram of the tones of the *Yancheng* dialect.



Appendix Figure 2 M2's z-score values diagram of the tones of the *Yancheng* dialect.



Appendix Figure 3 F1's z-score values diagram of the tones of the *Yancheng* dialect.



Appendix Figure 4 F2's z-score values diagram of the tones of the *Yancheng* dialect.